



Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

Network Routing and Switching

CCE-415

Theory Assignment – 02

Comprehensive IPv6 Architecture, Protocols, and Practical Implementation

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1. Introduction to IPv6 and Motivation

1. IPv6 is designed to be the successor to IPv4.
2. Depletion of IPv4 address space has been the motivating factor for moving to IPv6.
3. Projections show that all five RIRs will run out of IPv4 addresses between 2015 and 2020.
4. With an increasing Internet population, a limited IPv4 address space, issues with NAT and an Internet of things, the time has come to begin the transition to IPv6!
5. IPv4 has a theoretical maximum of 4.3 billion addresses, plus private addresses in combination with NAT.
6. IPv6 larger 128-bit address space provides for 340 undecillion addresses.
7. IPv6 fixes the limitations of IPv4 and includes additional enhancements, such as ICMPv6.

2. IPv6 Address Architecture & Representation Rules

128 bits represented in 8 groups of 4 hexadecimal digits separated by colons (e.g., 2001:0db8:0000:0000:0000:ff00:0042:8329).

• Shortening Rules:

1. Omit Leading Zeros

0db8 → db8, 0042 → 42.

2. Double Colon (::)

Replace a single contiguous block of consecutive all-zero groups with :: (applied only once per address).

► Shortened Form

2001:db8::ff00:42:8329/64.

- Prefix Notation is followed by a slash and prefix length indicating the network bits (standard end-host prefix is /64).

3. IPv6 Address Types and Scopes

Address Type	Prefix	Scope	Use Case
Global Unicast (GUA)	2000::/3	Global / Internet	Publicly routable host and server addressing
Unique Local (ULA)	fd00::/8	Local / Organization	Private routing within internal corporate networks
Link-Local (LLA)	fe80::/10	Single Link / Segment	Local communication, NDP, next-hop routing
Multicast	ff00::/8	Variable Scope	One-to-many communication (replaces broadcast)
Anycast	Same as GUA	Nearest Node	One-to-nearest service (DNS, load balancers)

4. Configuration Mechanisms: SLAAC, EUI-64, and DHCPv6

• SLAAC (Stateless Address Autoconfiguration):

- Host receives a /64 network prefix from router Router Advertisement (RA) messages.
- Host auto-generates its 64-bit Interface ID using EUI-64 or random generation without a DHCP server.

- **EUI-64 Calculation:**
 1. Take 48-bit MAC address (e.g., 00:11:22:33:44:55).
 2. Insert ff:fe in the middle → 0011:22ff:fe33:4455.
 3. Invert the 7th bit (Universal/Local bit) → 0211:22ff:fe33:4455.
- **DHCPv6:**
 - **Stateful DHCPv6:** Server centrally assigns IPv6 addresses, default gateway, and DNS servers (similar to IPv4 DHCP).
 - **Stateless DHCPv6:** Host uses SLAAC for its IP address and queries the DHCPv6 server only for auxiliary settings (e.g., DNS server, domain name).

5. Key Protocols: NDP and ICMPv6

- **Neighbor Discovery Protocol (NDP):**
 - Operates over ICMPv6 to replace IPv4 ARP, ICMP Router Discovery, and ICMP Redirect.
 - **Neighbor Solicitation (NS) / Neighbor Advertisement (NA):**
Resolves IPv6 addresses to Layer 2 MAC addresses and detects duplicate addresses (DAD).
 - **Router Solicitation (RS) / Router Advertisement (RA):**
Discovers default gateways and network prefixes.
- **ICMPv6:**
 - Handles diagnostic error reporting, path MTU discovery, multicast group management (MLD), and reachability testing (ping6).

6. Subnetting and Address Aggregation

- **Standard Subnet Size:** /64 is the standard prefix for all end-user network segments (2^{64} addresses per subnet).
- **Subnetting a /48 Allocation:**
 - Given ISP allocation: 2001:db8:1000::/48.
 - Subnet bits available: $64 - 48 = 16$ bits ($2^{16} = 65,536$ individual /64 subnets).
 - **Example Department Subnets:**
 - Engineering: 2001:db8:1000:0001::/64
 - Sales: 2001:db8:1000:0002::/64
 - Management: 2001:db8:1000:0003::/64
- **Address Aggregation:** Upstream routers summarize all 65,536 subnets into a single /48 route advertisement, keeping global routing tables compact.

7. Transition Mechanisms and Security

- **Transition Mechanisms:**
 - **Dual Stack:** Nodes run IPv4 and IPv6 protocol stacks concurrently.
 - **Tunneling (e.g., 6to4, GRE):** Encapsulates IPv6 packets inside IPv4 headers to traverse IPv4-only transit backbones.
 - **Translation (NAT64/DNS64):** Converts IPv6 packet headers to IPv4 headers for legacy system inter-communication.
- **Security & Privacy:**
 - **IPsec:** Natively supported in IPv6 for network-layer encryption and data authentication.
 - **Privacy Extensions (RFC 4941):** Generates randomized temporary interface identifiers to prevent tracking client devices across networks.

8. Practical Implementations and Troubleshooting

1. Basic Linux IPv6 Configuration & Verification:

```
sudo ip -6 addr add 2001:db8:1:1::10/64 dev eth0
ping6 -c 4 2001:db8:1:1::1
traceroute6 2001:db8:1:1::1
```

2. DHCPv6 Server Configuration (/etc/dhcp/dhcpd6.conf):

```
subnet6 2001:db8:1:2::/64 {
    range6 2001:db8:1:2::100 2001:db8:1:2::200;
    option dhcp6.name-servers 2001:db8:1:2::1;
    option dhcp6.domain-search "pstu.ac.bd";
}
```

3. SLAAC Troubleshooting Case Study:

- **Issue:** Hosts fail to auto-configure addresses via SLAAC.
- **Diagnosis:** Inspect Router Advertisement flags:
 - ManagedAddressConfiguration (M-flag = 1) forces hosts to query Stateful DHCPv6 instead of using SLAAC.
 - AutonomousAddressConfiguration (A-flag = 0) blocks SLAAC interface ID generation.
- **Fix:** Ensure RA advertises A-flag = 1 and appropriate prefix lifetimes.